X ColdShield

Physics-Aware OT Security Gateway

for Refrigeration and Cold-Chain Infrastructure

TECHNICAL DOSSIER & BUSINESS PLAN

France 2026 | Innovative Economic Project / French Tech Ecosystem

Laboratory Prototype - MVP 0.2.1

162	162	13	0
tests / Python 3.11	tests / Python 3.14	automated demo steps	failures, errors or skips

Founder & Technical Architect

Ramon M. Reis

Validated branch: codex/mvp-v0.2.1

Validated commit: 85d0c8ba2d9393ce923efb4a24cc582148df7e45

Public technical repository

github.com/undereis/x-coldshield-mvp

Scan to review the source, documentation and evidence.

October 2026

Confidential - incubation and innovation assessment

1. Executive Summary

THE PROBLEM AND THE X COLDSHIELD APPROACH

Modern food retail and cold-chain operations increasingly depend on connected refrigeration infrastructure. Many environments still include industrial controllers, gateways and maintenance systems designed primarily for operational availability rather than modern cybersecurity requirements.

Remote maintenance, interconnected store networks and legacy industrial communication can create situations in which an unauthorized or compromised command affects the physical operation of refrigeration equipment. In this context, a cyber incident can become an operational incident involving product preservation, equipment behavior and business continuity.

> **Core proposition**
>
> X ColdShield is a laboratory-stage Physics-Aware OT Security Gateway designed to add a security enforcement layer between external or maintenance networks and legacy refrigeration control systems. It evaluates the operational meaning of supported industrial commands and applies security policies based on expected physical operating constraints.

What the current prototype demonstrates

- Parsing and validation of a controlled subset of Modbus TCP requests.
- Explicit authorization of supported functions, unit IDs and synthetic registers.
- Configurable thermal policy enforcement before selected write commands are forwarded.
- Stateful defrost interlocking for defined laboratory scenarios.
- Time-limited maintenance windows with maintenance closed by default.
- Structured audit records that distinguish security decisions from upstream outcomes.
- Fail-closed behavior for invalid policy, unsupported traffic and selected failure conditions.

Responsible positioning

X ColdShield does not replace refrigeration controllers and is not presented as a finished industrial cybersecurity product. The present objective is to validate the technical concept in a reproducible laboratory environment, then progress through controlled industrial validation with refrigeration and OT security specialists in France.

> **Development principle**
>
> Laboratory validation → controlled industrial validation → commercial deployment. Each transition is evidence-gated and requires new acceptance criteria.

2. Core Technological Innovation

PHYSICS-AWARE OT SECURITY

The central technological hypothesis is that security decisions in Operational Technology should incorporate knowledge of the physical process being protected. An industrial command can be syntactically valid while still representing an operationally dangerous action.

Current technical mechanisms

Protocol-aware inspection

The MVP validates Modbus TCP framing and a controlled laboratory subset of Function Code 03 (Read Holding Registers) and Function Code 06 (Write Single Register). Unsupported or malformed requests are rejected.

Physics-aware thermal policy

Selected synthetic registers are associated with temperature setpoints. The demonstration can permit a legitimate -22 C maintenance command while blocking an attempted +50 C modification. These thresholds are demonstration rules, not validated physical limits.

Stateful defrost interlock

Security decisions can depend on prior operational state, not only the current packet. The prototype blocks a defined conflicting defrost scenario within its simulated model.

Controlled maintenance

Write access is closed by default and can be opened only for a bounded maintenance period. Version 0.2.1 serializes administrative changes and handles selected persistence failures conservatively.

Structured audit

Events include request IDs, policy context, decision reasons and communication outcomes. The audit is designed for laboratory traceability and is not claimed to be immutable.

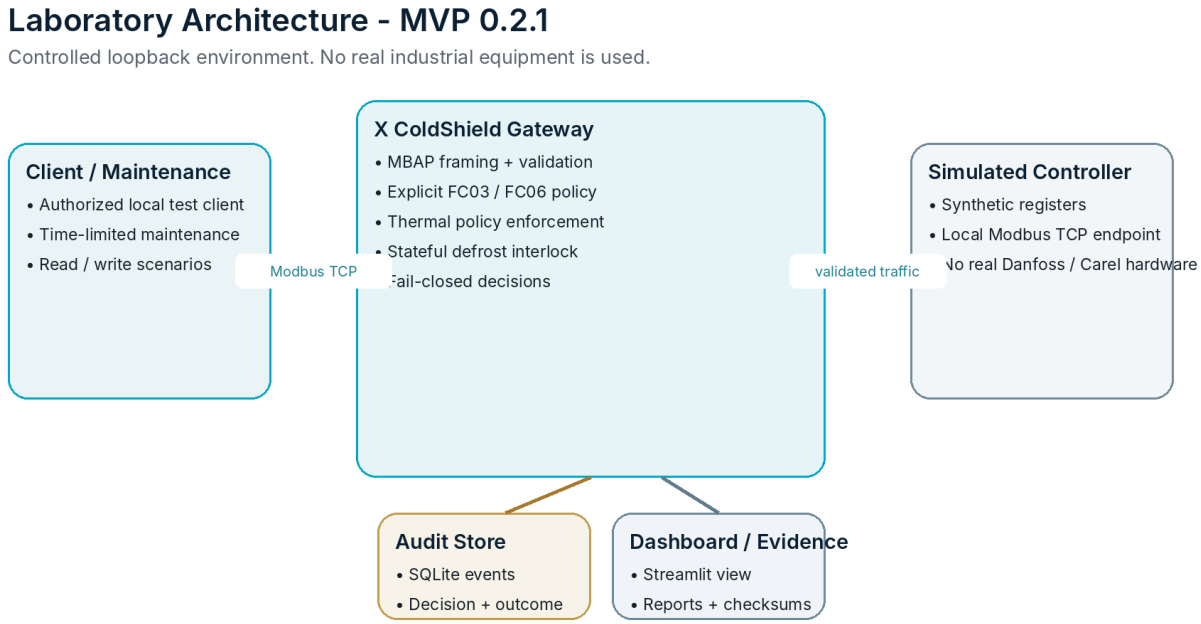


Figure 1. Current laboratory architecture. All traffic is local and simulated.

3. Validated Laboratory MVP - Version 0.2.1

REPRODUCIBLE SOFTWARE EVIDENCE

A functional laboratory prototype is publicly available for technical review. The current validation target is branch codex/mvp-v0.2.1 at commit 85d0c8ba2d9393ce923efb4a24cc582148df7e45.

Repository: github.com/undereis/x-coldshield-mvp **CI run:** GitHub Actions #38742283406

Evidence	Status
Python 3.11 automated suite	**162 tests - Passed**
Python 3.14 automated suite	**162 tests - Passed**
Failures, errors and skips	**0 - Passed**
Automated demonstration	**13 steps - Completed**
CI evidence and checksums	**Available in repository**

Key modules in the current architecture

Module	Role
coldshield/protocol.py	Modbus TCP framing and message validation
coldshield/gateway.py	Central policy enforcement and outcome handling
coldshield/server.py	Bounded client sessions and controlled shutdown
coldshield/maintenance.py	Time-limited maintenance authorization and serialization
coldshield/storage.py	Persistent laboratory audit/state with retention controls
coldshield/simulator.py	Synthetic refrigeration controller for isolated testing
scripts/demo.py	Automated demonstration and evidence generation

Current limitations

The prototype is not industrially certified and does not claim complete Modbus compatibility, BACnet support, authenticated technician identity, immutable logs, validated thermodynamic safety limits, NIS 2 certification or compatibility with real Danfoss / Carel installations. No current evidence was produced against operational refrigeration infrastructure.

4. Market Opportunity & Business Model

INITIAL FOCUS: FOOD RETAIL AND COLD-CHAIN OPERATIONS

X ColdShield is initially aimed at environments where refrigeration is essential to business continuity. Potential customer segments include supermarkets and hypermarkets, cold-storage facilities, refrigerated logistics operators, refrigeration maintenance companies, system integrators and OT cybersecurity integrators.

Why the problem matters commercially

- Legacy and long-lived OT assets can remain in service far longer than standard IT systems.
- Remote maintenance and enterprise connectivity can enlarge the attack surface around physical processes.
- Replacing an entire refrigeration control estate can be expensive and operationally disruptive.
- A security layer that can be introduced without immediately replacing the controller estate may offer a lower-friction path to improved control and visibility.

Proposed commercial model

> **B2B edge + recurring software**
>
> A future deployment could combine a hardened edge gateway with recurring software services for policy management, audit visualization, alerting, fleet management, reporting and SOC integration. Integration and professional services could support installation, controller mapping and OT security assessment.

Market assumptions still to validate

- Willingness to pay and final subscription pricing per site.
- Installation and support cost for different refrigeration estates.
- Procurement cycles and pilot requirements of French retail operators.
- Preferred routes to market: direct sales, refrigeration integrators or OT security partners.
- Commercial value of physics-aware controls compared with existing OT security platforms.

Regulatory context

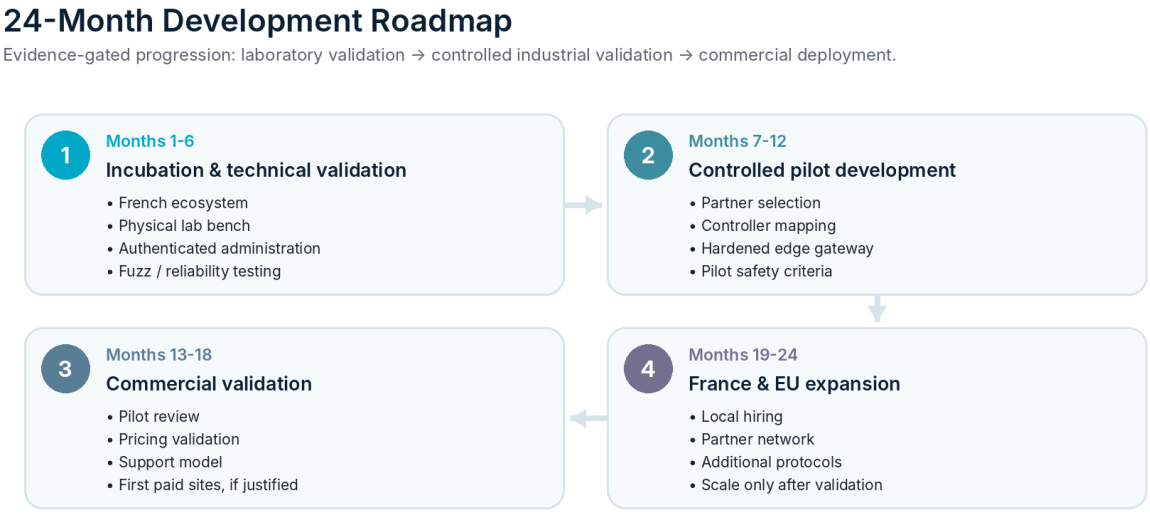
The project is being developed in a European environment with increasing emphasis on cybersecurity risk management, supply-chain security and protection of important digital systems. X ColdShield may eventually support technical controls useful to organizations operating under frameworks such as the EU NIS 2 Directive; however, the project does not claim NIS 2 certification or automatic regulatory compliance.

> **Target-market references**
>
> Major groups such as Carrefour, E.Leclerc, Intermarche, Systeme U and Auchan illustrate the scale and sophistication of the French food retail market. They are target-market examples only; no partnership or customer relationship is claimed.

5. Why France & 24-Month Roadmap

BUILDING THE NEXT VALIDATION STAGE INSIDE THE FRENCH ECOSYSTEM

France is the preferred location for development because it combines a significant food retail sector, industrial and refrigeration engineering capabilities, a strong cybersecurity ecosystem, technology incubators and access to European partners. The intention is to develop X ColdShield from France with local technical partners, not simply relocate an existing business.



Technical priorities before any industrial pilot

- Authenticated administration and stronger technician identity.
- Fuzz testing, sustained-load testing and recovery testing.
- Controller-specific register mapping validated by refrigeration specialists.
- Physical operating limits derived from equipment documentation and refrigeration engineering.
- Independent OT security review and explicit acceptance criteria for a controlled test bench.
- Broader protocol support only where justified by partner needs and test evidence.

Pilot gate

Any test with real refrigeration equipment would require a separate technical authorization, risk assessment, validated register map and rollback / safety plan. The current release does not authorize control of industrial equipment.

6. Founder Profile & Current Project Status

TECHNICAL ENTREPRENEURSHIP AND INFRASTRUCTURE EXPERIENCE

Ramon M. Reis - Founder & Technical Architect

Ramon M. Reis is a Brazilian technology entrepreneur and infrastructure professional with more than two decades of practical experience across telecommunications, network infrastructure and technology operations. His background combines hands-on technical work with the operation of telecommunications and Internet service businesses in Brazil.

Relevant experience includes Linux systems, IP networking, routing and switching, BGP / OSPF, network security, firewall architecture, fiber-optic telecommunications, ISP infrastructure, wireless networks, monitoring, automation, databases and cloud technologies. He completed higher education in Information Systems in Brazil in 2022.

X ColdShield represents the convergence of this networking and infrastructure experience with a new focus on Operational Technology and cyber-physical security. As founder, Ramon is responsible for product concept, technical architecture, prototype development direction, security requirements, validation strategy, business positioning and engagement with the French ecosystem.

Current project status - October 2026

Evidence	Status
Concept definition	**Completed**
Laboratory MVP	**Completed**
Independent technical review of 0.2	**Completed**
MVP 0.2.1	**Completed**
Automated suite - Python 3.11 / 3.14	**162 / 162 - Passed**
Automated demonstration	**13 steps - Completed**
Public technical repository	**Available**
Incubator outreach	**Initiated**
Industrial equipment validation	**Not yet performed**
Commercial pilot / revenue validation	**Not yet performed**

Founder objective in France

The objective is to establish and develop X ColdShield in France and progressively build a French-based technology company focused on OT cybersecurity, with local validation partners and a responsible path from laboratory evidence to controlled industrial deployment.

7. Project Vision, Support Sought & Evidence

FROM LABORATORY VALIDATION TO CONTROLLED INDUSTRIAL VALIDATION

The long-term vision is to develop a security layer capable of understanding not only who is communicating with an industrial system, but also whether the requested physical action makes operational sense. The current refrigeration prototype provides a controlled environment in which this concept can be explored, tested and challenged.

Support sought from the French ecosystem

- Incubation and technical mentoring.
- Refrigeration engineering expertise and controlled laboratory access.
- OT cybersecurity review and pilot-security design.
- Introductions to refrigeration integrators and potential early adopters.
- Business-model validation, regulatory guidance and access to innovation networks.
- Support defining the evidence required before a controlled industrial pilot.

Evidence links

Repository: https://github.com/undereis/x-coldshield-mvp

MVP 0.2.1 branch: codex/mvp-v0.2.1

CI evidence: GitHub Actions run 38742283406

> **Responsible technical statement**
>
> X ColdShield is presented as a technically implemented and reproducibly tested laboratory prototype exploring a physics-aware approach to OT cybersecurity. It is not presented as a certified industrial safety system, a NIS 2 certification mechanism, or a production-ready controller for refrigeration equipment.

Next milestone

The next challenge is to validate the concept together with refrigeration specialists, industrial cybersecurity professionals and French technology partners. The project seeks to convert reproducible software evidence into a controlled physical test-bench program, with explicit safety criteria, before any commercial deployment.

Laboratory validation → Controlled industrial validation → Commercial deployment

X ColdShield | Physics-Aware OT Security Gateway | France 2026